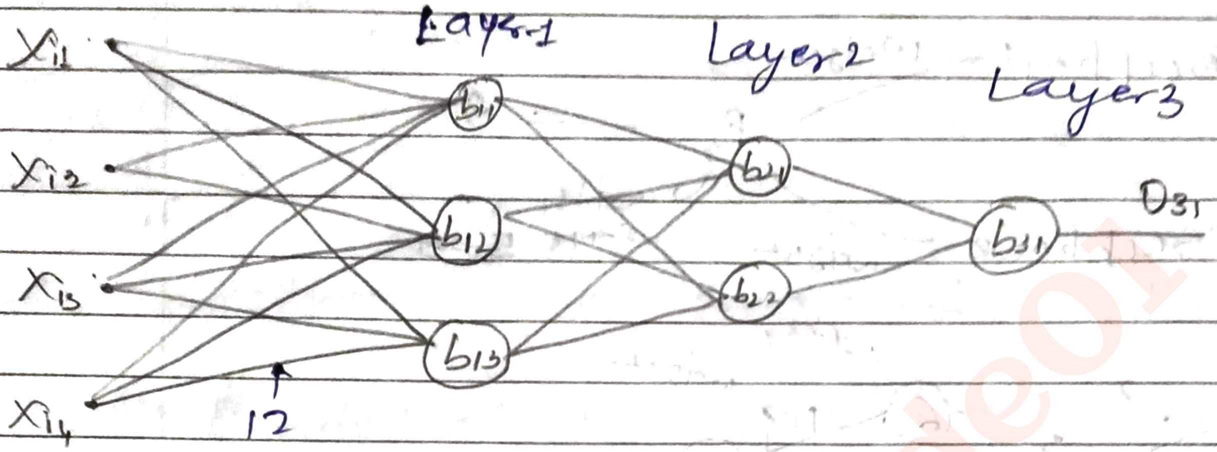
SdS

Day-10

→ Forward Propagation

- The movement of data from input layer through the hidden layer to the output layer in a neural network.



No. of trainable parameters $= 12+3+6+2+2+1$
$= 26$

dataset

cgpa	iq	10th	12th	placed
7.2	72	69	81	1
8.1	92	75	76	0

prediction → $\sigma(W^T x + b)$

Layer 1

$$\begin{bmatrix} w^1_{11} & w^1_{12} & w^1_{13} & \cancel{w^1_{14}} \\ w^1_{21} & w^1_{22} & w^1_{23} & \cancel{w^1_{24}} \\ w^1_{31} & w^1_{32} & w^1_{33} & \cancel{w^1_{34}} \\ w^1_{41} & w^1_{42} & w^1_{34} & \cancel{w^1_{44}} \end{bmatrix}^T \begin{bmatrix} X_{i1} \\ X_{i2} \\ X_{i3} \\ X_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$4\times3 \xrightarrow{\text{Transpose}} 3\times4 \times 4\times1 \qquad 3\times1$

$3\times1 \qquad 3\times1$

Shivamcode01

SdS

$$\Rightarrow \begin{bmatrix} w_{11}^1 x_{i11} + w_{21}^1 x_{i12} + w_{31} x_{i13} + w_{41} x_{i14} \\ w_{21} x_{i2} + \cdot \quad \cdot + \cdot \quad \cdot + \\ w_{31} x_{i3} + \quad + \quad + w_{43} x_{i4} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$\sigma\left(\begin{bmatrix} \quad & + b_{11} \\ \quad & + 42 \\ \quad & b + b_{13} \end{bmatrix} \right)$$

$$\Rightarrow \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix}$$

Layer-2

$$\begin{bmatrix} w_{11}^2 & w_{12}^2 & \boxed{w_{13}^2} \\ w_{21}^2 & w_{22}^2 & \\ w_{31}^2 & w_{23}^2 & \end{bmatrix}^T \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$3 \times 2 \xrightarrow{T} 2 \times 3 \qquad 2 \times 1$$

$$\Rightarrow \sigma\left(\begin{bmatrix} w_{11}^2 O_{11} + w_{21}^2 O_{12} + w_{31}^2 O_{13} + b_{21} \\ w_{12}^2 O_{11} + w_{22}^2 O_{12} + w_{23}^2 O_{13} + b_{22} \end{bmatrix} \right)$$

$$\Rightarrow \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix}$$

Layer-3

$$\begin{bmatrix} w_{11}^3 \\ w_{21}^3 \end{bmatrix}^T \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$$2 \times 1 \xrightarrow{T} 1 \times 2 \qquad 1 \times 1$$

ShivamXcode01

$$\sigma\left(\left[w^{3}_{11}O_{21} + w^{3}_{21}O_{22} + b_{31}\right]\right) = \hat{y}_1 = O_{31}$$

$$a^{[1]} = \sigma\left(a^{[0]}W^{[1]} + b^{[1]}\right)$$

$$a^{[2]} = \sigma\left(a^{[1]}W^{[2]} + b^{[2]}\right)$$

$$a^{[3]} = \sigma\left(a^{[2]}W^{[3]} + b^{[3]}\right)$$

final Expression :

$$\underbrace{\sigma\Big(\underbrace{\sigma\big(\big(\underbrace{\sigma(a^{[0]}W^{[1]}+b^{[1]})}_{a^{[1]}}\big)W^{[2]}+b^{[2]}\big)}_{a^{[2]}}\cancel{a}W^{[3]}+b^{[3]}\Big)}_{a^{[3]} = \text{output}}$$

↪ What happend inside neural network

ShivamxCode01